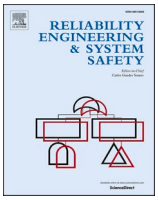
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Resilience Metric of Equipment System: Theory, Measurement and Sensitivity Analysis

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ABSTRACT

This paper analyzes the equipment system resilience metric by employing reliability engineering parameters. It proposes a hybrid resilience metric by analyzing the applicability of two existing methods —- deterministic metric and probabilistic metric. Boundary analysis verifies the correctness under special circumstances. Based on the reliability engineering parameters of equipment, the concrete representation of hybrid measurement method is displayed in the usual "fault – repair" mode of equipment system. Then, under the given fault mode and repair strategy, and according to the diversity of fault and repair situation of complex equipment system after disturbance, the resilience metric of complex equipment system is proposed to represent the inherent properties of equipment. Moreover, the influence of reliability engineering parameters on the resilience metric of the equipment system is given in the sensitivity analysis. No matter in equipment design or improvement stage, the failure rate is a parameter which is far more important than the others, and importance of the others is related to the specific system. This study provides quantitative indexes for the performance of the whole resilience process from fault to repair after disturbance, which can offer the support for optimizations of equipment design and improvement.

1. Introduction

When an equipment breaks down after disturbance, which of the following three situations can reflect the equipment's best disturbance rejection ability? (1) The equipment's performance is completely lost when suffering from failure but recovering quickly. (2) The equipment's performance is partially lost when suffering from failure but recovering slowly. (3) The equipment's performance is lost or reduced with probability when suffering from failure, and then recovering. Which factor does contribute to the different results and how much do they affect the results in the above situations? These problems cannot be solved easily by traditional reliability engineering(RE) theory, which has been widely researched on reliability [1–5], robustness [6–11], restorability [12–15], et al.. Characters in traditional RE theory only take out some one point of the process of equipment failure, but problems still exist in the whole process. After the introduction of resilience to ecology [16] and other areas, resilience engineering can solve the above problems very well, especially in the field of infrastructure system [17–21], which is used to describe the ability of a system to withstand, absorb, and recover from disturbance.

Leveson [22] proposed a new accident model in 2004, which laid the groundwork of resilience engineering and attracted more and more researches of resilience in engineering area. By finding that resilience was confused with robustness, fault tolerance, flexibility, survivability, and agility in traditional RE in many researches, Henry [23] pointed out that the resilience was relied on time. He divided resilience process into three states – initial state, post-disturbance state and post-recovery state – and two processes – performance degradation process and performance recovery process, which were combined with related conceptions of RE. Then this thought was widely accepted in engineering area [24]. Patriarca [25] thought that resilience engineering was the extension of reliability by analyzing related papers of resilience. Aven [26] thought that not only disturbance events, states of the system, and the result of disturbance, but also uncertainties should be considered, when researching performance of system in resilience engineering. The U.S. Department of Defense(DoD) [27] proposed an engineered resilient system strategy in 2011, which pointed out that system must have the ability to repel/resist/absorb disruptions, to recover from disruptions, and to adapt to new or changed conditions. Goerger [28] proposed four factors of resilient defense system, adding the ability to operate

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effectively in uncertain environments, based on these three abilities and the report of Military Operations Research Society. Naghshbandi [29] also pointed out the significance of uncertainty and interconnectedness in complex engineering system resilience.

The equipment in engineering area is a complex system in nature, which is composed of components assembled in terms of function and structure. For equipment system, the problem on resilience research is which research needs to be carried out and how to research. Similar researches are mainly concentrated on power grid resilience at present [30–32]. If power grid is viewed as an equipment, the reduction of grid capacity by overload or failure of components can be regarded as the equipment performance degradation by disturbance or failure. And the change of grid's structure and recovery strategy are similar to equipment improvement and repair strategy. The difference is that the grid has the ability of self-adaptability but the equipment requires human intervention to adjust. All above indicate that it is very important to carry out the research on the resilience of equipment system.

The equipment system researched in this manuscript is one that is used for specific purpose in engineering field, which requires a lot of parameters to describe the ability of equipment system in traditional RE area such as failure rate (FR), mean time to repair (MTTR), mean time between failure (MTBF), repair rate (RR), and so on. The equipment system can be divided into simple system and complex system in terms of structural complexity. The simple system is an equipment system without subsystem, and is usually the equipment minimum replaceable unit. Therefore, its failure mode is single and can be described by a set of RE parameters. However, the complex equipment is an equipment system containing subsystems, among which the RE parameters of each subsystem constitute the RE parameters of the whole complex system. In this paper, the resilience metric is mainly carried out under the assumption that the equipment system is a simple system. For complex systems, the resilience metric can be obtained based on simple system, considering the diversity of performance changes with failure, as well as the various possibilities for repair. Only one aspect of the complex equipment system resilience is researched in this paper.

Although these studies on RE can hardly solve the problems faced by resilience engineering, the relationship between RE and resilience engineering – the contents of RE correspond to each stage of the resilience process – may be useful for resilience research in the engineering area. As a highly mature research area, a lot of researches have been done to RE on the whole process from failure to recovery, and there are many typical parameters to describe the performance of the system. These parameters may be helpful to the study of resilience engineering.

The RE parameters, such as FR, MTTR, MTBF, RR, et al., are inherent parameters of a system, which may be affected by environmental, personal and other factors in practice. It describes a specific ability at different times in resilience process. And what we concern about is whether it can be used to describe the inherent resilience of equipment system or not. The capability of the system in this environment based on the inherent resilience can be described quantitatively by using the parameters that are affected by environmental, personal and other factors.

Determining how to measure the resilience of equipment system by using RE parameters, what is the relationship between resilience and the parameters in traditional RE, and determining how to use resilience to solve the problem at the beginning of this paper are key problems to be solved first of equipment system resilience, which are also the research contents of this paper.

The content of this paper is arranged as follows. In section 1, the present research, the significance, and main contents of the research on the equipment system resilience are analyzed. In section 2, the common metrics of resilience are described. In section 3, a hybrid metric of resilience is proposed, and the boundary of the metric is analyzed. In section 4, methods of optimizing system capability based on resilience in equipment design and equipment improvement are analyzed respectively.

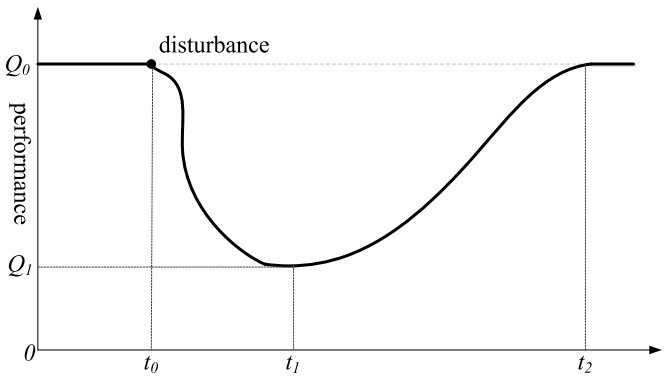


Fig. 1. Schematic figure of resilience process

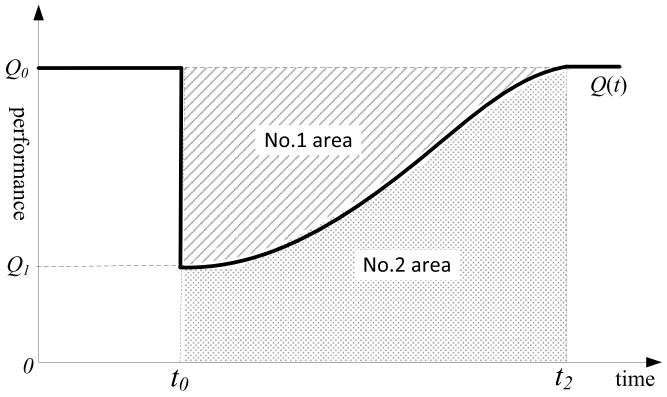


Fig. 2. Schematic figure of deterministic metric

2. Literature review

In the overall design and performance analysis of equipment system, the equipment needs to obtain a high resilience to ensure that it can still complete the mission under disturbance. So, the quantitative assessment of resilience is significant to the research on resilience engineering of equipment system. Due to the diversity of systems and environments, scholars lay different emphases on measuring resilience. There is no uniform standardized method for measuring resilience, which can be mainly divided into two categories [33]: deterministic metrics and probabilistic metrics, although some scholars use Delphi method to measure resilience [34].

2.1. Deterministic Metrics

Deterministic metric method is mainly adopted to measure the system's resilience according to the change of performance in resilience process, which is shown in Fig. 1. Bruneau [18] measured resilience by the loss performance of the system from system disturbance to performance recovery:

$$R = \int_{t_0}^{t_2} [1 - Q(t)]dt \tag{1}$$

where t_0 is the moment when the disturbance occurs, and t_2 is the moment when the system performance recovers completely, and $Q(t)$ is the performance curve of the system.

Similarly, Cimellaro [35] measured resilience by the residual performance of the system from system disturbance to performance recovery:

$$R = \int_{t_0}^{t_2} Q(t)dt \tag{2}$$

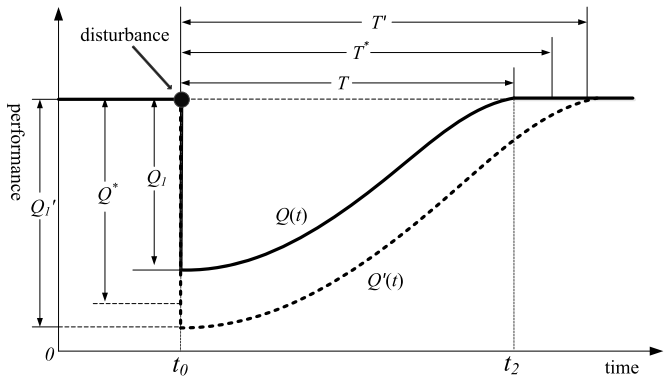


Fig. 3. Schematic figure of probabilistic metric

Reed [36], Zobel [37,38], Ouyang, [39,40] et al. have done a lot of researches on the deterministic metric of resilience and given the corresponding equations. These methods mainly measured the resilience based on the changes of the system's performance after the disturbance. The core idea of these methods was to consider the accumulation of the system's performance or performance loss during the whole resilience process. Area 1 and area 2 in Fig. 2 show the loss and residual performances respectively. Chao Zhang [41] proposed another kind of deterministic metric in network resilience, which measured resilience by network velocity, based on defining the resilience of node and edge.

2.2. Probabilistic Metrics

Probabilistic metric method is mainly used to measure the system's resilience by considering the randomness of the system's performance degradation under disturbance and of the recovery time. By introducing the resilience triangle, Bruneau [18] measured resilience in terms of the probability of system performance degradation and recovery time reaching a certain threshold. Chang and Shinozuka [42] defined resilience as the probability of performance loss and recovery time not exceeding the given maximum value, shown as:

$$R_i(Q^*, T^*) = P(Q < Q^*, T < T^* | i) \tag{3}$$

In this equation Q and Q^* are the actual maximum performance loss and the threshold respectively. Similarly, T and T^* are the actual maximum recovery time and the threshold respectively. i is the disturbance event. Based on this, a resilience metric under random disturbance is:

$$R = \sum_i R_i(Q^*, T^*)P(i) \tag{4}$$

where $P(i)$ is the probability of event i occurring.

There are also many scholars considering not only the threshold, but also the uncertainty during entire resilience process, whose core is regarding disturbance, performance loss and performance recovery as random events and using random variables to describe resilience directly [36,39,40,42]. Fig. 3 plots the different resilience processes $Q(t)$ and $Q^{'}(t)$, where the disturbance is random.

A special kind of probabilistic metrics is Bayesian network method, which employs the probability to describe the resilience capability of system based on Bayesian theory and conditional probability theory [43]. This method is widely used in measuring supply chain resilience in recent years [44,45], for example in the field of waterway transportation [46], urban infrastructure [47], manufacturing [48], and energy [49].

In conclusion, there are a lot of different measuring methods from different pointcuts to resilience, which can be divided into two categories, deterministic metric and probabilistic metric, focusing on performance's changing process and uncertainty respectively.

3. Methods

3.1. Hybrid Metrics

Deterministic metric and probabilistic metric study resilience from different viewpoints respectively. Deterministic metric mainly considers the process of system performance change, while probabilistic metric mainly considers the uncertainty. Both of them take the change of system performance into account, of which research objects are mainly in the field of traditional engineering, not taking into the effect of strong disturbance caused by complex system and complex environment, which are common in equipment system. So, when measuring resilience of equipment system, the following factors must be considered:

(1) The change of the environment affects the failure rate and disturbance rate directly, for the uncertainty of the environment equipment works.
(2) In the process of equipment running, there may be a different intensity of failure or disturbance, which leads to a different reduction of performance.
(3) The speed and quality of the performance recovery are related with the performance of the equipment itself and the environment.
(4) The equipment system has continuity in running time, and its performance change affects the ability to complete the specified function.

So, we must use different probabilities in different environments, different changes of performance on different disturbance and maintenance states, different time consumptions of performance changing process, and the whole process from the occurrence of disturbance to the end of maintenance.

Generally, it needs to consider the whole process of probability, performance degree, and recovery when measuring resilience of equipment system. The deterministic metric pays no attention to the uncertainty of the formation of the disturbance and the influence of the disturbance on the system. Also, the probabilistic metric pays no attention to the continuity of the process of the disturbance occurring to the performance recovery. All above cannot measure the resilience of the equipment system comprehensively.

Considering the features of these two kinds of methods, a series of resilience process $Q_i(t)$ are calculated in resilience metric, and each of these $Q_i(t)$ is a possible situation, which has its own probability of occurrence $P(i)$. Considering a cluster of possible performance changes curves of the system after disturbance, which represented by $\mathbf{Q}(t)$, where $Q_i(t) \in \mathbf{Q}(t)$, a hybrid metric can be calculated as

$$R = \sum_{Q_i \in \mathbf{Q}} P(i) \int Q_i(t)dt \tag{5}$$

The performance change curve is determined by disturbance and recovery events, performance change and time of the whole process. However, it is usually difficult to obtain the performance change curve directly. But for equipment system, the performance variation can be obtained through its characteristics.

Considering the continuity and randomness of the resilience process, the system resilience can be expressed as the sum of several stages of the whole process. For one of these $\mathbf{Q}(t)$, it can be separated into two parts, q_1 and q_2, representing failure and maintenance processes respectively. q_1 is the change curve of equipment performance from disturbance to the beginning of repair, and q_2 is from the beginning of repair to the end of repair, both related to disturbance. Then, for equipment system, the hybrid metric can be calculated as

$$R_1 = P_F(x)P_{Fq}(q_1|x)\int_{t_0}^{t_1} q_1(t)dt \tag{6}$$

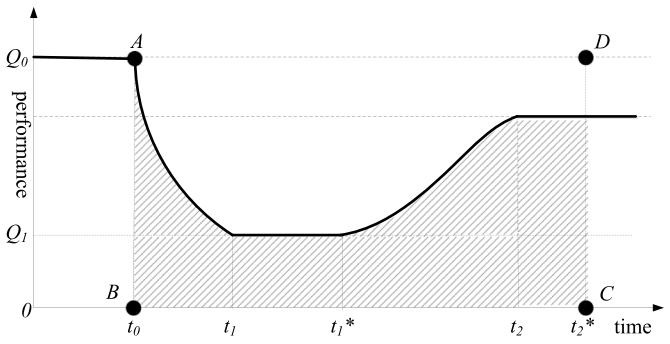


Fig. 4. Hybrid metric of resilience

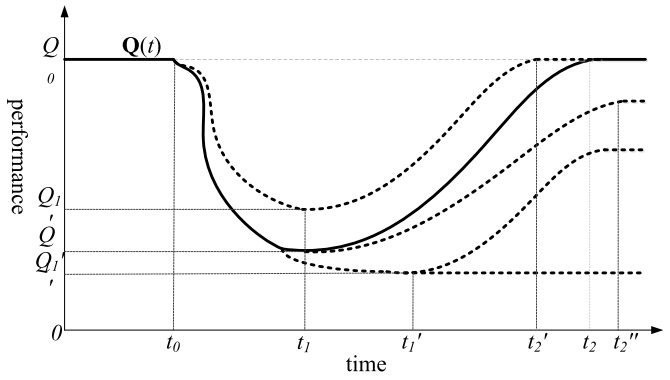


Fig. 5. A cluster of resilience processes

$$R_2 = P_F(x)P_{Fq}(q_1|x)P_{Rq}(q_2|x,q_1)\int_{t_1}^{t_1^*} q_1(t_1)dt \tag{7}$$

$$R_3 = P_R(x)P_{Rq}(q_2|x,q_1)\int_{t_1^*}^{t_2} q_2(t)dt \tag{8}$$

$$R_4 = P_R(x)P_{Rq}(q_2|x,q_1)\int_{t_2}^{t_2^*} q_2(t_2)dt \tag{9}$$

Where X is the set of disturbance events. $P_F(x)$ is the probability of the occurring disturbance x. $P_{Fq}(q_1|x)$ is the conditional probability with the system's performance degradation curve $q_1(t)$ under the disturbance x. $P_R(x)$ is the repair rate under the disturbance x. $P_{Rq}(q_2|x,q_1)$ is the conditional probability of system's performance recovery curve under the disturbance x and the performance reduction curve $q_1(t)$. t_1, t_1^*,t_2, and t_2^* are respectively the ending moment of performance degradation, the beginning moment of recovery, the ending moment of recovery, and the maximum recovery time allowed by the system under the disturbance x, the performance degradation curve $q_1(t)$, and the performance recovery curve $q_2(t)$. $R_i(i=1,\cdots,4)$ are the integral of performance change curve over the process of performance degradation, the process of performance remaining, the process of performance recovery and the process after recovery respectively. The metric of resilience can be expressed as

$$R = \frac{\sum_{x\in X}\sum_{q_1\in Q_1}\sum_{q_2\in Q_2}(R_1+R_2+R_3+R_4)}{t_2^*-t_0} \tag{10}$$

where R is the resilience of the equipment system, which is in the range of $[0,1]$, Q_1 is the set of the performance degradation curve, and Q_2 is the set of the performance recovery curve.

As shown in Fig. 4, Eq. (10) reflects the expectation of the ratio between the residual performance integral under random disturbance – shaded area – and the performance integral without performance degradation - rectangular ABCD area.

3.2. Relationship

Hybrid metric is the development and combination of the deterministic and the probabilistic, which is equivalent to these two in certain cases.

As demonstrated in Fig. 5, for a cluster of resilience processes $\mathbf{Q}(t)$, the deterministic only considers one of these processes and the probabilistic only considers the probability of different processes. However, the hybrid is a comprehensive result of deterministic metric for a cluster of resilience processes $\mathbf{Q}(t)$ that may occur by considering all the possible resilience processes and analyzing the probability of occurrence of each process comprehensively.

The expression of the resilience metric shows that lower probability of disturbance, less performance degradation after disturbance, higher probability of recovery, shorter the recovery time, and more performance recovery result in a higher resilience.

With the Deterministic

The deterministic does not consider the probability of the resilience process, but all the events occur during one certain process. When the probability of disturbance, performance degradation, performance recovery and other events are particular value 1 in the hybrid metric, equal to $P_F(x)P_{Fq}(q_1|x)P_R(x)P_{Rq}(q_2|x,q_1) = 1$, which means all random events must happen, Eq. (10) can be simplified as

$$R = \frac{\int_{t_0}^{t_1} q_1(t)dt + \int_{t_1}^{t_1^*} q_1(t_1)dt + \int_{t_1^*}^{t_2} q_2(t)dt + \int_{t_2}^{t_2^*} q_2(t_2)dt}{t_2^*-t_0} \tag{11}$$

This description is like Eq. (2), which uses residual performance to measure the resilience. It is a kind of deterministic metric, whose value is the ratio of the area of the shaded part in Fig. 2 to the area of the rectangle ABCD.

With the Probabilistic

The probabilistic does not consider the resilience itself, but the probability during the process. When the performance degradation and performance recovery process ignore the time factor, which means only the post-disturbance performance q_1 and the post-recovery performance q_2 are considered, Eq. (10) can be simplified as

$$R = \sum_{x\in X}\sum_{q_1\in Q_1}\sum_{q_2\in Q_2}\left[P_F(x)P_{Fq}(q_1|x)p_1(x,q_1) + P_R(x)P_{Rq}(q_2|x,q_1)(p_2(x,q_1,q_2) - p_1(x,q_1))\right] \tag{12}$$

The metric equation is the expectation of performance retention capability, which is a kind of expectation representation in probabilistic metric. When no disturbance can cause any influence, equal to $p_1(x,q_1) = 1$, the resilience is $R = 1$, which means the system has the maximum resilience. When the performance loses due to disturbance, equal to $p_1(x, q_1) = 0$, the resilience metric depends on the recovery rate and recovery degree of the performance.

3.3. Boundary Analysis

For general equipment system, the performance decreases after disturbance, and then recovers to a certain level after repair. The resilience is from 0 to 1. What will happen in a special case? According to the definition of resilience, if an equipment fails and cannot be repaired under disturbance, it has the lowest resilience; and if it has no change when suffering from disturbance, it has the highest resilience. These boundary analyses are aiming at verifying the correctness in these two special cases.

As 0 probability leads to 0 result, we just consider the different

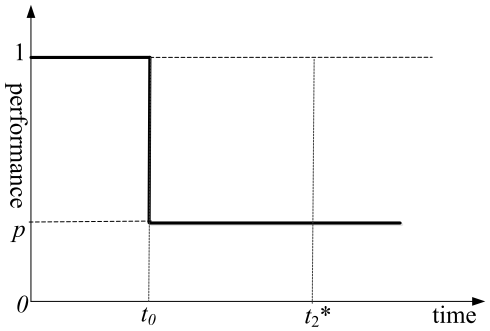


Fig. 6. Resilience process of unrecoverable disturbance

special performance changing processes when disturbance occurs in boundary analysis. In this section, we set all probability to value 1 or 0 in the calculating process. The different performance changing processes mainly describe that whether the performance changes and how to change when disturbance occurs. For equipment system, it is normal that performance degrades immediately when disturbance occurs (failure) and recovers immediately when repairment occurs (repair), and it is also normal that there is a period after the failure and before repair.

Boundary 1: unrecoverable disturbance

It is the case that the performance of the equipment degrades by suffering from unrecoverable disturbance. When the equipment suffers from a disturbance, the performance degrades to $p(p < 1)$ immediately and cannot be recovered, leading to the system broken-down. The process of this case is shown in Fig. 6 and the performance degradation curve can be expressed as

$$q_1(t) = \begin{cases} 1 & t = t_0 \\ p & t > t_0 \end{cases} \tag{13}$$

the performance recovery curve can be expressed as

$$q_2(t) = p \tag{14}$$

and the resilience of the equipment system can be expressed as

$$R_1 = P_F(x)P_{Fq}(q_1|x)\int_{t_0}^{t_1} q_1(t)dt = P_F(x)P_{Fq}(q_1|x)\cdot 0 = 0 \tag{15}$$

$$\begin{aligned} R_2 &= P_F(x)P_{Fq}(q_1|x)P_{Rq}(q_2|x,q_1)\int_{t_1}^{t_1^*} q_1(t_1)dt \\ &= P_F(x)P_{Fq}(q_1|x)P_{Rq}(q_2|x,q_1)p\left(t_1^* - t_1\right) \end{aligned} \tag{16}$$

$$R_3 = P_R(x)P_{Rq}(q_2|x,q_1)\int_{t_1^*}^{t_2} q_2(t)dt = P_R(x)P_{Rq}(q_2|x,q_1)p\left(t_2 - t_1^*\right) \tag{17}$$

$$R_4 = P_R(x)P_{Rq}(q_2|x,q_1)\int_{t_2}^{t_2^*} q_2(t_2)dt = P_R(x)P_{Rq}(q_2|x,q_1)p\left(t_2^* - t_2\right) \tag{18}$$

$$\begin{aligned} R &= \lim_{t_1 \to t_0^+} \frac{\sum_{x\in X}\sum_{q_1\in Q_1}\sum_{q_2\in Q_2}(R_1 + R_2 + R_3 + R_4)}{t_2^* - t_0} \\ &= \lim_{t_1 \to t_0^+} \frac{p\left(t_1^* - t_1\right) + p\left(t_2 - t_1^*\right) + p\left(t_2^* - t_2\right)}{t_2^* - t_0} \\ &= \lim_{t_1 \to t_0^+} \frac{p\left(t_2^* - t_1\right)}{t_2^* - t_0} = p \end{aligned} \tag{19}$$

Specially, when $p = 0$, which means that the equipment system has no ability to withstand disturbance and be repaired, the value of resilience is 0.

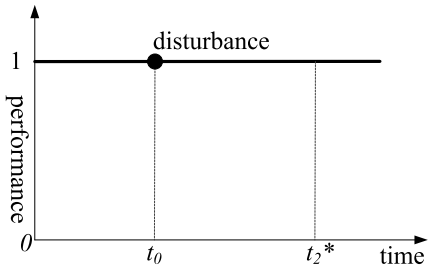


Fig. 7. Resilience process of unaffected disturbance

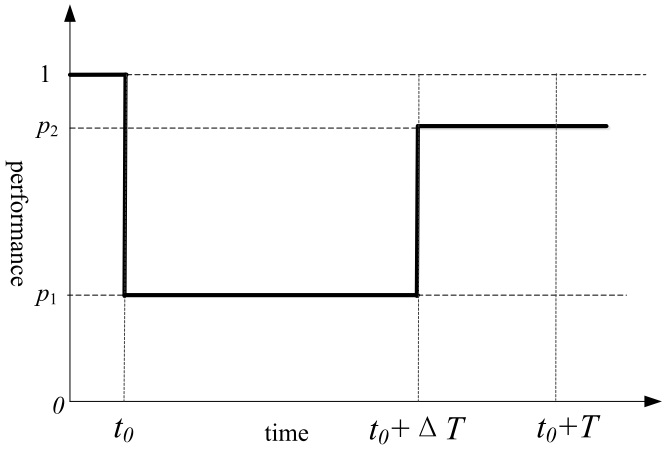


Fig. 8. Resilience process of recovery after disturbance

Boundary 2: unaffected disturbance

It is the case that the performance of the equipment remains unchanged after disturbance. When the equipment suffers from a disturbance, the performance is still 1 and the system does not need to be repaired, so $P_R(x) = 0$, $q_1(t) = 1$, and $q_2(t) = 1$, which means that the system can work normally. The process of this case is displayed in Fig. 7 and the resilience of the equipment system can be expressed as

$$R_1 = P_F(x)P_{Fq}(q_1|x)\int_{t_0}^{t_1} q_1(t)dt = P_F(x)P_{Fq}(q_1|x)(t_1 - t_0) \tag{20}$$

$$\begin{aligned} R_2 &= P_F(x)P_{Fq}(q_1|x)P_{Rq}(q_2|x,q_1)\int_{t_1}^{t_1^*} q_1(t_1)dt \\ &= P_F(x)P_{Fq}(q_1|x)P_{Rq}(q_2|x,q_1)\left(t_1^* - t_1\right) \end{aligned} \tag{21}$$

$$R_3 = P_R(x)P_{Rq}(q_2|x,q_1)\int_{t_1^*}^{t_2} q_2(t)dt = P_R(x)P_{Rq}(q_2|x,q_1)\left(t_2 - t_1^*\right) \tag{22}$$

$$R_4 = P_R(x)P_{Rq}(q_2|x,q_1)\int_{t_2}^{t_2^*} q_2(t_2)dt = P_R(x)P_{Rq}(q_2|x,q_1)\left(t_2^* - t_2\right) \tag{23}$$

$$\begin{aligned} R &= \frac{\sum_{x\in X}\sum_{q_1\in Q_1}\sum_{q_2\in Q_2}(R_1 + R_2 + R_3 + R_4)}{t_2^* - t_0} \\ &= \frac{(t_1 - t_0) + \left(t_1^* - t_1\right) + \left(t_2 - t_1^*\right) + \left(t_2^* - t_2\right)}{t_2^* - t_0} = 1 \end{aligned} \tag{24}$$

This process indicates that when the system has infinite ability of resistance to disturbances, the resilience is 1.

3.4. Common case: repair after failure

Commonly, when an equipment system failure, its performance may degrade to value p_1 immediately and recover to value p_2 after a period

ΔT, which means that the time of system failure state is ΔT. The maximum recovery time allowed by the system is T. As shown in Fig. 8, the performance degradation curve can be expressed as

$$q_1(t) = \begin{cases} 1 & t = t_0 \\ p_1 & t > t_0 \end{cases} \tag{25}$$

the performance recovery curve can be expressed as

$$q_2(t) = \begin{cases} p_1 & t = t_0 + \Delta T \\ p_2 & t > t_0 + \Delta T \end{cases} \tag{26}$$

and the resilience of the equipment system is

$$R_1 = P_F(x)P_{Fq}(q_1|x)\int_{t_0}^{t_1} q_1(t)dt = P_F(x)P_{Fq}(q_1|x)\cdot 0 = 0 \tag{27}$$

$$\begin{aligned} R_2 &= P_F(x)P_{Fq}(q_1|x)P_{Rq}(q_2|x,q_1)\int_{t_1}^{t_1^*} q_1(t_1)dt \\ &= P_F(x)P_{Fq}(q_1|x)P_{Rq}(q_2|x,q_1)p_1\left(t_1^* - t_1\right) \end{aligned} \tag{28}$$

$$R_3 = P_R(x)P_{Rq}(q_2|x,q_1)\int_{t_1^*}^{t_2} q_2(t)dt = P_R(x)P_{Rq}(q_2|x,q_1)\cdot 0 = 0 \tag{29}$$

$$R_4 = P_R(x)P_{Rq}(q_2|x,q_1)\int_{t_2}^{t_2^*} q_2(t_2)dt = P_R(x)P_{Rq}(q_2|x,q_1)p_2\left(t_2^* - t_2\right) \tag{30}$$

$$\begin{aligned} R &= \frac{\sum_{x\in X}\sum_{q_1\in Q_1}\sum_{q_2\in Q_2}(R_1 + R_2 + R_3 + R_4)}{t_2^* - t_0} \\ &= \frac{p_1\left(t_1^* - t_1\right) + p_2\left(t_2^* - t_2\right)}{t_2^* - t_0} = \frac{p_1\Delta T + p_2(T - \Delta T)}{T} \\ &= p_2 + \frac{\Delta T}{T}(p_1 - p_2) \end{aligned} \tag{31}$$

As $p_1 - p_2 \le 0$, for this process, more quickly the system recovers, more resilient the system becomes. When $\Delta T = 0$, this case is the same as boundary 1, and the post-disturbance performance is p_2. When $p_2 = p_1$, this case is also the same as boundary 1, and the post-disturbance performance is p_1.

For an equipment system without subsystem, we can assume that the system will suffer from only one disturbance, which means that there is only one performance degradation event and one performance recovery event. Then, there are two situations of the system after disturbance – failure and normal. The probabilities $P_F(x)$ and $P_F(p_1|x)$ can be combined into P_F, which means that the probability of disturbance occurs and leads performance to degrade to p_1. Similarly, there are two situations of the system after repair – repair and disrepair. The probabilities $P_R(x)$ and $P_R(p_2|x,p_1)$ can be combined into P_R, which means that the probability of performance recovers to p_2. The resilience metric can be expressed as

$$R = (1 - P_F) + P_F(P_R(\alpha p_1 + (1-\alpha)p_2) + (1 - P_R)p_1) \tag{32}$$

Where $\alpha = \Delta T/T$, which represents the failure time ratio; the value range of p_2 is in the interval $[p_1, 1]$; and the value range of all other parameters is in the interval $[0, 1]$. The first part of this equation is the situation that system is normal, and the second part is the situation of failure. Also in the second part, the front half is the situation of repair, the second is disrepair.

Among the above parameters, P_F can be equivalent to the failure rate in traditional RE, P_R can be equivalent to the repair rate, ΔT can be equivalent to mean time to repair (MTTR), and T can be equivalent to the sum of MTTR and mean time between failure (MTBF). The degree of performance after degradation q_1 and performance after repair q_2 are related to system robustness and recovery.

3.5. Complex system

For complex system combined with subsystems, the changing of the performance depends on the reaction of each subsystem under disturbance. Pan Xing considered three situations in resilience process of the system when analysis system resilience [50], and taken weighted network as the object to study the recovery strategy of the system [51]. The essence of redundancy is to reduce the failure rate of system. The essence of failure-free is to reduce the failure rate of one specified subsystem to 0. The essence of quick repairing is to considering some subsystems repairing but the others not for reducing the whole repairing time of system. These are similar to the conditions of complex system when subjected to disturbance. The redundancy and failure-free in system reflect the reliability and robustness, which influence the parameters P_F and q_1. The quick repairing reflects the recovery, which influence the parameters P_R and q_2. The above influence the parameters MTTR and MTBF together. Therefore, for complex systems, we must consider the different cases of the variation of these parameters.

In the cases where the parameters are possible, complex systems may have three conditions of performance loss (Q=0), degradation (0<Q<1) and no impact (Q=1), when the system is subjected to disturbance resulting in subsystem failure. In this case, for describing the whole complex system, the RE parameters of each subsystem must be used. It means that from simple equipment system to complex equipment system, we should describe it by adjusting a single-parameter set to a multi-parameter one. For example, in a complex system composed of two subsystems, the nonfailure rate is $(1 - P_{F1})(1 - P_{F2})$, and the failure rate is represented by $P_{F1}(1 - P_{F2})$ (subsystem 1 failure only), $(1 - P_{F1})P_{F2}$ (subsystem 2 failure only), and $P_{F1}P_{F2}$ (both subsystems failure). Moreover, the performance is not completely lost during the failure, which is determined by the failure subsystem.

In this case, we can still use Eq. (5) to measure the resilience of a complex system, but the difference from a simple system is that $Q_i(t) \in \mathbf{Q}(t)$ represents one of the changes. This situation contains all possibilities that no subsystem fails after the disturbance, at least one subsystem is repaired after the failure, and none of the subsystems can be repaired after the failure.

If the complex system is composed of n subsystems and the number of fault subsystems is k, there are $\sum_{k=1}^{n} C_n^k$ kinds of fault situations in the complex system when at least one subsystem fails; if i of the k fault subsystems are repaired, there are $\sum_{i=1}^{k} C_k^i$ kinds of repair situations. The failure and repair situations and their probabilities corresponding to different performance degradation curves and performance recovery curves and their conditional probabilities are in Eq. (6)~(9).

But there are some problems that, for the same equipment system under the same disturbance, different fault response modes and maintenance strategies will lead to completely different performance change curves. Because these strategies are completely different for different operators and managers, it is difficult to give an exact performance change curve to reflect the inherent performance of the equipment. However, it is certainly feasible to extend this measurement from simple systems to complex systems theoretically.

It is feasible to measure the resilience of a complex equipment system on this basis that assuming that all the failed subsystems fail at the same time after disturbance, and maintenance resources can guarantee the simultaneous repair of all the failed subsystems.

Let $\mathbf{X}$ represents a cluster of faulty subsystems, X is a cluster of failed subsystems, $q_1(X)$ represents the performance of the whole complex system when the subsystem in set X fails, $\mathbf{Y}$ represents a cluster of repaired subsystems after a set of subsystems fail, and $Y \in \mathbf{Y}$, $Y \subset X$. For the set of repaired subsystems, each subsystem is numbered according to the repair order. If $q_2(X|i)$ is the performance after the repair of the subsystem no. I; $q_2(X|0) = q_1(X)$ is the performance after failure; T_i is the MTTR of subsystem no. I; $T_0 = 0$ is the initial moment; and $T = \max(\mathrm{MTTR}_i + \mathrm{MTBF}_i)$ is the total time of the resilience process. Then,

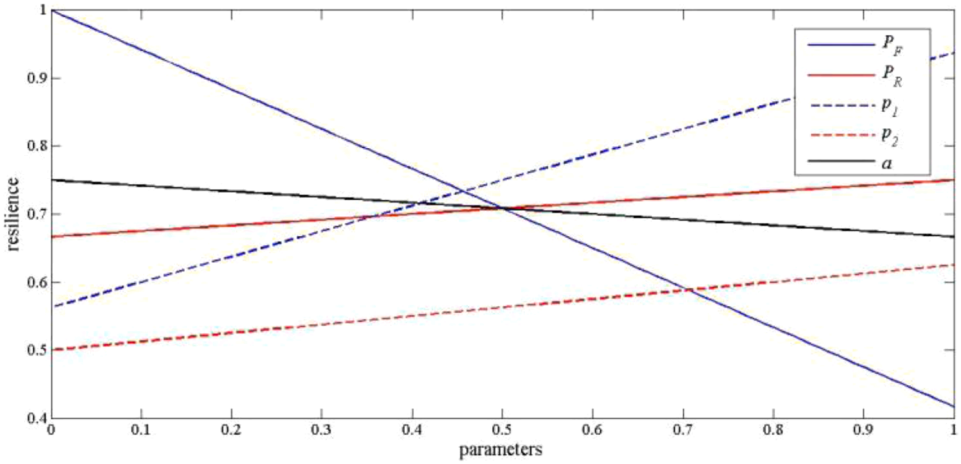


Fig. 9. The change of resilience's expectation

the resilience process of complex system can be expressed as

$$R_1 = \prod_{i=1}^{n}(1 - P_{F_i}) \tag{33}$$

$$R_2 = \prod_{i\in X}(1 - P_{R_i})\cdot q_1(X)T \tag{34}$$

$$R_3 = \sum_{Y}\prod_{i\in Y}P_{R_i}\prod_{i\notin Y}(1 - P_{R_i})\sum_{i\in Y}[q_2(X|i-1)(T_i - T_{i-1})] \tag{35}$$

Where R_1 represents the nonfailure situation, R_2 represents the unrepairable situation after failure, and R_3 represents the repairable situation after failure. Since the failure rate of subsystems in set X is $\prod_{i\in X}P_{F_i}\prod_{i\notin X}(1 - P_{F_i})$, the resilience of the complex system can be expressed as

$$R = R_1 + \sum_{X}\prod_{i\in X}P_{F_i}\prod_{i\notin X}(1 - P_{F_i})(R_2 + R_3)\Big/T \tag{36}$$

When the number of subsystems in a complex system is one, the system is simplified to a simple system, and there is only one set of fail-repair modes. The above formula will also be simplified to Eq. (6) to Eq. (10). In the different resilience strategies, the P_F of subsystem may be 0 (protected) or $P_F{\cdot}P_F$ (redundancy), and the P_R of subsystem may be close to 1 (quick repairing), in which other value of these parameters depends on the system itself and the resilience strategy together.

Hybrid metric describes the resilience process by considering the degradation process of performance and the probability totally, which can reflect the ability of system for resisting disturbance well.

4. Sensitivity Analysis

In the process of equipment system design or improvement, determining how to reasonably design or allocate the RE parameters of equipment to keep its high resilience is the problem we are concerned about in the research of the resilience engineering. Without considering the environmental and human factors, whether simple or complex equipment systems, the inherent resilience of equipment system is determined by the following factors, P_F, P_R, q_1, q_2 and α, which are defined in section 3. Sensitivity analysis is to study the influence of the above parameters on resilience. This paper mainly analyzes from two aspects of equipment system design and improvement. For parameter variation, there are parameters being undetermined and parameters being determined, respectively.

4.1. Undetermined Parameters

For equipment system design, all parameters are undetermined. The essence that which set of parameters is optimal is to analyze the influence of the elements of resilience and give the principle of determining parameters.

When one of five above parameters is fixed, we take the resilience's expectation of the remaining parameters varying within the range of values as R in the case of the determined value. Fig. 9 shows the change

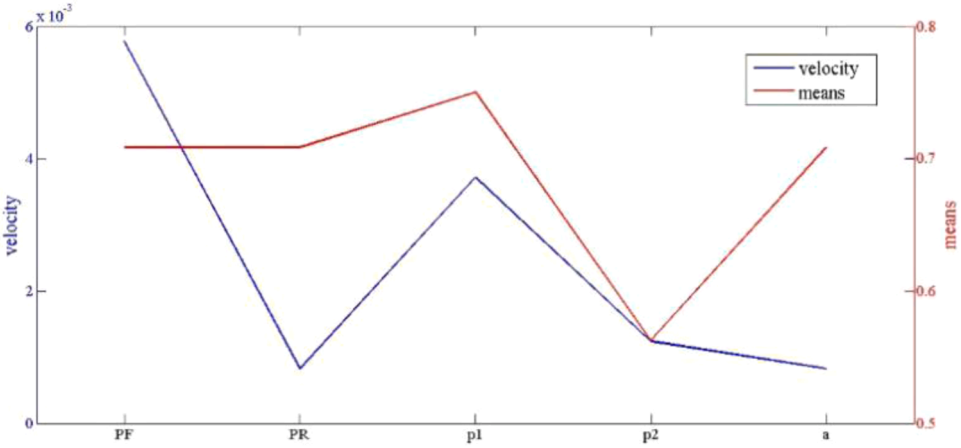


Fig. 10. The changing velocity and means of resilience's expectation

Table 1
Initial value

Parameter	P_F	P_R	p_1	p_2	a
Value	0.1	0.8	0.2	0.8	0.3

of the resilience's expectation when above five parameters change within the value range. The changing velocity of resilience's expectation and the means of resilience's expectation corresponding to different parameters are plotted in Fig. 10.

It can be concluded that the resilience's expectation of P_F changes the fastest, and P_R and α change the slowest. For the means of resilience's expectation, p_2 is significantly lower than other parameters.

When designing the equipment system, the parameter with the greatest influence on resilience should be given priority, and the parameter with the least influence should be considered at last. Parameter optimization should be considered in the following order:

(1) It needs to make the failure rate low enough, which includes both the probability of disturbance and the probability of performance degradation. Reducing the probability of disturbance involves the ability of prediction and prevention of disturbance, whose essence is to improve the system's reliability. Reducing the probability of performance degradation involves the ability to absorb the disturbance, whose essence is to improve the system's robustness.
(2) It needs to make the degradation of system performance low enough, whose essence is to improve the system's robustness.
(3) It needs to make the repair rate high enough, whose essence is to improve the system's restorability.
(4) It needs to make the failure time ratio low enough. Reducing the failure time ratio is reducing MTTR and increasing MTBF, whose essence is also to improve the system's recovery.
(5) It needs to make the system performance recovery high enough, whose essence is also to improve the system's restorability.

4.2. Determined Parameters

For equipment system improvement, all parameters are determined. The essence that how to adjust parameters optimally is to analyze the influence of the elements of resilience and give the principle of adjusting parameters.

When parameters are determined, what is considered in optimization is no longer the change of resilience's expectation, but the change of resilience itself caused by parameter adjustment under this set of parameters. Table 1 shows an example of the initial parameters of an engineering system. In this case, Fig. 11 shows the change of resilience when one of these parameters' changes around the initial value and the remaining parameters remain unchanged. The parameter variation is adjusted step by step according to 10% of distance from the original parameter to its boundary. Fig. 12 displays the changing velocity of

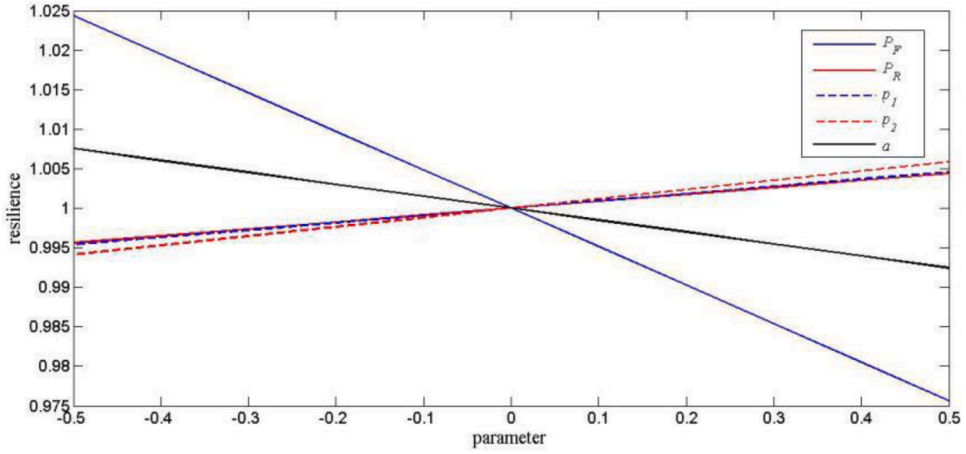


Fig. 11. The change of resilience

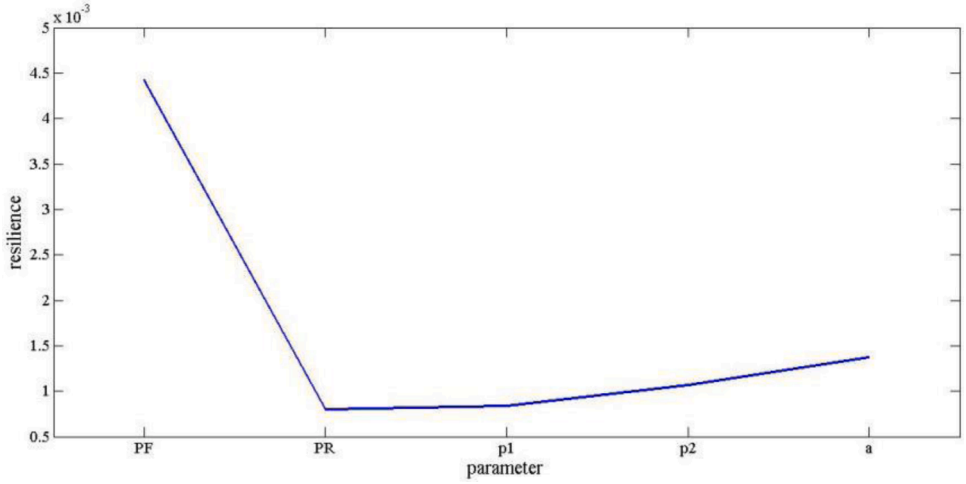


Fig. 12. The changing velocity of resilience

resilience.

When optimizing the equipment system, the parameters with the greatest influence on resilience should be given priority, and the least influenced should be considered at last. Parameter optimization should be considered in the following order:

(1) It needs to reduce the failure rate, which is usually decided by the structural design of the system and its adaptability to the environment.
(2) It needs to reduce the failure time ratio, which is usually decided by transport strategy and recovery strategy.
(3) It needs to improve the performance recovery degree, which is usually decided by equipment repairability.
(4) It needs to reduce the performance degradation, which is usually decided by the structural design of the system.
(5) It needs to improve the repair rate, which is usually decided by repair capability.

When optimizing a certain equipment system, the importance of the parameters is related to the current resilience state of the system, which is related to the parameters' initial value.

Above all, for enhancing the resilience of a system, reducing the failure rate of system of each subsystem is the most important, and then the performance retaining, repair rate and the repair time. In other words, the reliability is the most important part of the resilience.

5. Conclusion and future research

We analyze the resilience metrics and present a hybrid metric by considering the key elements of RE, which can be regarded as deterministic metric or probabilistic metric in some special situations. By analyzing the boundary of the hybrid metric, it is concluded that the resilience metric results under special circumstances accord with the objective law. Based on the boundary analysis, we come up with the resilience metric of simple equipment system and complex equipment system respectively. Then, the sensitivity is researched by discussing the relationship between parameters that constitute the resilience and parameters in traditional RE, such as FR, MTTR, MTBF, et al. Finally, the design of the system and the measures to deal with the failure of the system are analyzed, and the optimization principles are given.

The parameters that constitute the resilience in this research are all parameters in the field of traditional RE or elements involved in equipment maintenance process. So, this method has a good inheritance to the traditional RE research methods. The influence of the above parameters on the system resilience is not isolated. In practical equipment system design and maintenance decision, the inherent parameters of system, operating environment, the difficulty, and cost of parameter adjustment all should be taken into consideration.

In this paper, the inherent resilience of the equipment system is measured based on the inherent parameters of the equipment – RE parameters. As in RE, the equipment system will be affected by the environment and people who operate it. Determining how to measure the resilience of equipment in different environments, determining how to reflect the failure mode and maintenance strategy in the resilience metric, and whether the task recombination has an impact on the resilience of complex equipment system may be the possible topics for the further research on the resilience of equipment system.

Declaration of Competing Interest

All authors of this manuscript have directly participated in planning, execution, and analysis of this study.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ress.2021.107889.

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